Warm and Cold Cycles

Conventional Chronology, UNIQUANT and Orbital Frequency Patterns

A reproducible comparison over the last 420,000 years

Exploratory analysis of the UNIQUANT framework

6 November 2026
Version 1.0; not peer reviewed

Abstract

This calculation places the archived Vostok isotope-derived temperature reconstruction beside long-term Earth-orbit variables from La2004. It compares their frequency patterns and tests two distinct meanings of "Quant time." An exact conversion of the published chronology to the reference unit $q = 75/1012$ seconds leaves normalized frequency content unchanged when the frequency unit is converted consistently. Separately, extrapolating the manuscript's quadratic model-year function produces small numerical age changes on this interval. These are unvalidated clock scenarios, not revised dates. The figures show the actual reconstructed climate series and computed orbital variations; they do not infer a new cosmic energy field or add incommensurate amplitudes as energy. Source files, numerical tables and the analysis script accompany the report.

Direct answer. Warm/cold intervals and orbital frequency patterns can be displayed in a common framework. The exact Quant coordinate supplies a different time scale label. It does not by itself move an ice age or create a new frequency.

What is compared. The climate curve is a local Antarctic reconstruction, not global mean temperature. The conventional axis is its published GT4 chronology, with ages before 1950 (BP). The orbital curves describe eccentricity, obliquity, climatic precession and a selected top-of-atmosphere insolation quantity. These have different units and physical roles.

Scope. The analysis concerns multi-millennial to glacial cycles. Daily, monthly and annual oscillations cannot be recovered from an archive sampled at intervals of decades to millennia. The earlier short planetary trajectory comparison is not extrapolated into this 420,000-year calculation.

1 Climate, orbital variations and frequency images

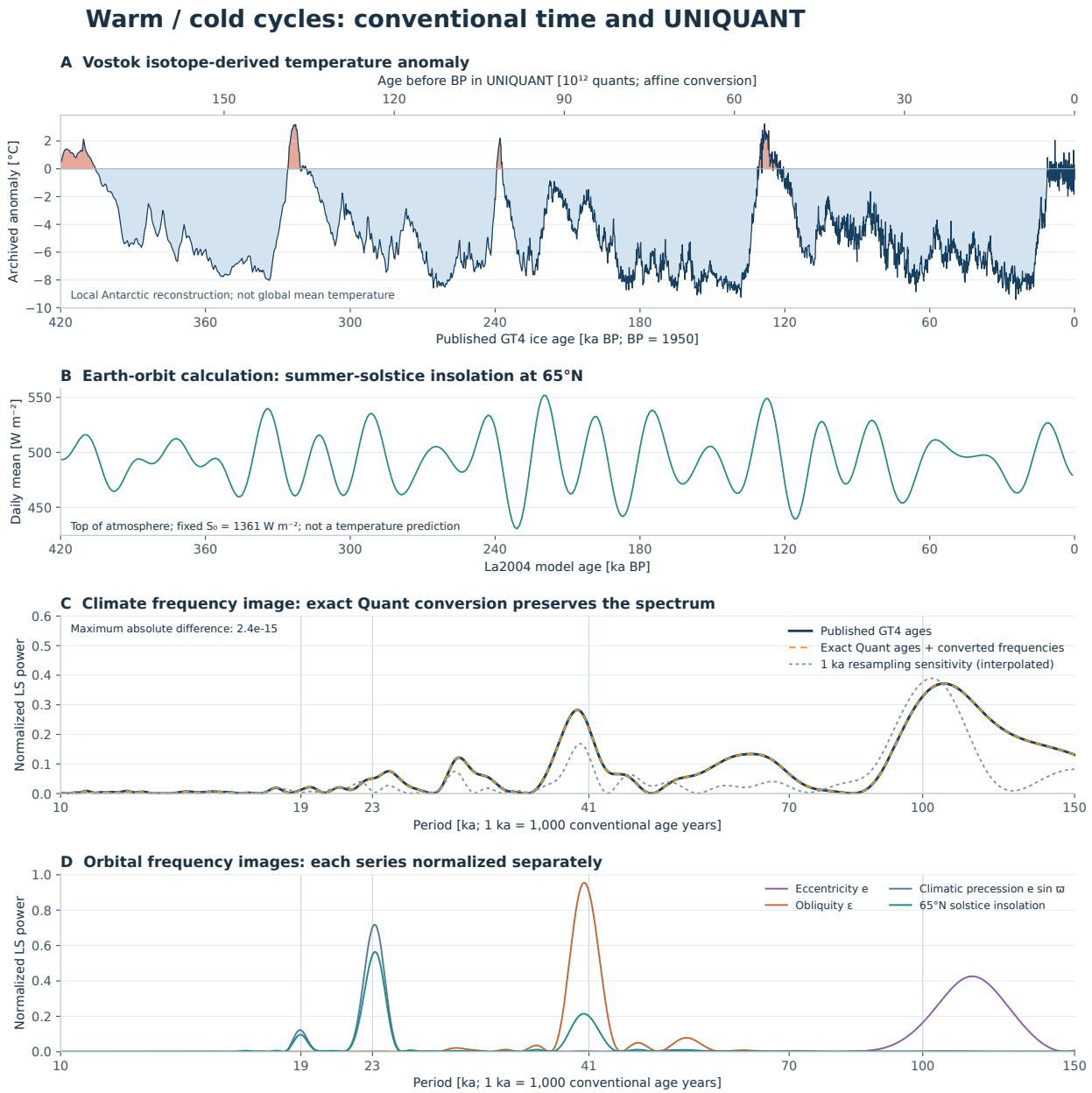


Figure 1: The archived Vostok reconstruction and the long-term La2004 orbital solution, with a conventional age axis and its exactly rescaled Quant-age coordinate. Periodograms show improvement of individual sinusoidal fits, not energy densities. Period guide lines mark familiar comparison scales, not imposed physical clock frequencies. All spectra are calculated from the displayed data series. Source: NOAA/Petit et al. and IMCCE/Laskar et al.; processing in the accompanying script.

2 Data and spectral calculation

2.1 A published chronology is retained

The NOAA Vostok archive supplies depth, GT4 ice age, deuterium isotope values and a derived temperature anomaly [1, 2]. The archive's temperature values are used unchanged and labelled "Vostok δD-derived temperature anomaly (archived reconstruction)." No conversion into global temperature is made. Ice age is used for the ice isotope series; gas age is a separate quantity. The selected interval ends above the disturbed bottom section discussed in the original study. The analysis metadata list the exact retained range and count.

GT4 combines ice-flow/accumulation modelling with adopted control ages. The temperature reconstruction also includes an ocean-isotope correction. The original study's chronology assessments were of order thousands of years; they are historical model assessments, not an uncertainty covariance supplied for every sample [2]. No new independent age observations are added here.

2.2 Actual long-term orbital inputs

La2004 supplies time t_{2000} in kyr from J2000, eccentricity e, obliquity ϵ and heliocentric perihelion longitude ϖ [3, 4]. We use the age convention

$$a_{\mathrm{BP}}\,[\mathrm{ka}] = -t_{2000} - 0.05, \qquad p = e\sin\varpi.$$

For the chosen daily-mean northern summer-solstice quantity at latitude $\phi = 65^\circ$, let $\lambda = \pi/2$, $\delta = \epsilon$, and $H_0 = \arccos(-\tan\phi\tan\delta)$. Then

$$I_{65} = \frac{S_0}{\pi}\frac{[1 - e\cos(\lambda - \varpi)]^2}{(1-e^2)^2}\left[H_0\sin\phi\sin\delta + \cos\phi\cos\delta\sin H_0\right]. \tag{1}$$

$S_0 = 1361\,\mathrm{W\,m^{-2}}$ is a fixed chosen normalization. The minus sign uses Earth's heliocentric perihelion longitude. This quantity is not a global annual heat input or a direct prediction of Antarctic temperature.

2.3 What the frequency image measures

At every trial frequency f, fit a floating offset and a sine/cosine pair:

$$y_i = c_f + A_f\cos(2\pi f a_i) + B_f\sin(2\pi f a_i) + r_i, \qquad P(f) = 1 - \frac{\sum_i r_i^2}{\sum_i (y_i - \bar{y})^2}. \tag{2}$$

This is an equal-weight, floating-mean normalized Lomb–Scargle calculation. The climate samples remain at their uneven archived ages; interpolation is not needed for the primary spectrum. Dense age intervals receive more weight. An additional 1-ka interpolation check tests this sampling choice; it adds no independent observations. Each $P(f)$ is a separate sample-weighted fit improvement. Values at different frequencies are not additive energy fractions. No linear trend is removed. The fixed search spans periods 10–150 ka. The finite interval, sampling, chronology and climate response limit interpretation; a finely sampled frequency grid is not a precise physical age measurement.

3 Numerical comparison and the exact Quant coordinate

Series	Selected peak periods [ka]	$P(f)$
Vostok reconstruction	105.8; 39.8; 24.0	0.372; 0.283; 0.075
Eccentricity e	114.1	0.426
Obliquity ϵ	40.5	0.956
Climatic precession	23.1; 19.0	0.718; 0.122
65°N solstice insolation	23.2; 40.4; 19.0	0.564; 0.214; 0.097

Selected maxima, not an exhaustive peak catalogue. The climate entries are the maxima in the predefined 70–130, 35–50 and 17–26 ka bands. Orbital entries are the strongest separate local maxima in the analysis range.

Retained inputs: 3,306 Vostok samples, ages 0–419,682 years BP, and 420 orbital samples, ages 0.95–419.95 ka BP. These are nearby but not identical endpoints. No endpoint extrapolation is used in the primary periodograms.

Sampling check. Linear interpolation at 1-ka intervals gives corresponding band maxima near 102.4, 40.0, 22.3 ka. This is a processing sensitivity on the same archive, not new evidence or a better age model. The sampling choice changes the detailed spectrum much more than the trial clock curvatures do.

Exact Quant check. Maximum absolute difference of normalized powers: 2.44×10^{-15}. The direct least-squares cross-check agrees to floating-point precision.

The listed peaks are descriptive maxima on the specified grid. They are not statistical significance estimates, exact astronomical constants, or proof of causal coupling. A record of about 420 ka contains only about four 100-ka cycles. A 405-ka component is not claimed as an independently resolved cycle in this calculation. The source data and chronology can change the fine structure of the shorter-period bands.

3.1 Why an exact time-unit conversion preserves the spectrum

The reference step in the current UNIQUANT manuscript is

$$q = \frac{75}{1012}\,\mathrm{s}, \qquad Q = \frac{t - t_{jj}}{q}. \tag{3}$$

It is a defined coordinate grid, not an experimentally detected fundamental oscillation or quantum transition duration [5]. With the conventional Julian-year duration $Y_J = 31\,557\,600$ s, the conversion is exactly

$$\frac{Y_J}{q} = 425\,817\,216, \qquad K = 425\,817\,216\,000 \quad \text{Quants per ka.} \tag{4}$$

For the plots we use the positive Quant age before BP, $Q_{\mathrm{age}} = (t_{\mathrm{BP}} - t)/q = Ka_{\mathrm{BP}}$, rather than the signed jj coordinate. The jj anchor changes an origin; the positive age convention also reverses direction. Neither alters the available frequency information.

If $Q_{\mathrm{age}} = Ka$ and $f_Q = f/K$, then

$$2\pi f_Q Q_{\mathrm{age}} = 2\pi f a. \tag{5}$$

The design columns and fitted residuals are identical. An arbitrary additional origin shift rotates the sine/cosine coefficient pair and leaves its fitted amplitude and normalized power invariant. The script verifies the unit-conversion identity numerically. An unnormalized spectral density can change numerical scale with its units; that is why this comparison uses the same normalized fit definition.

4 The quadratic year function: a separate sensitivity scenario

The current manuscript also defines a model-year function

$$s(n) = q\left[Bn + \frac{\alpha n - bn^2}{529}\right] = An - cn^2, \quad (B, \alpha, b) = (425806080, 703993, 11), \tag{6}$$

with $A = q(B + \alpha/529)$ and $c = 11q/529$. Its coefficients are definitions in the manuscript; palaeoclimate calibration over hundreds of thousands of years is not established [5].

For illustration only, extend the stated polynomial to real-valued model-year indices and set a nominal BP model-year index $n_0 = -79.52886$. This fixes the scenario convention relative to the future jj label; it is not a fitted or independently measured ancient clock index. For conventional age $u = 1000a$ in years,

$$s(n_0) - s(n_0 - u) = s'(n_0)u + cu^2. \tag{7}$$

Two calculations must be distinguished:

$$u_{\text{literal}} = \frac{s'(n_0)}{Y_J}u + \frac{c}{Y_J}u^2, \qquad \text{literal polynomial-to-seconds scenario,} \tag{8}$$

$$u_{\text{local}} = \frac{s(n_0) - s(n_0 - u)}{s'(n_0)} = u + \beta u^2, \qquad \beta = \frac{c}{s'(n_0)}. \tag{9}$$

The local version is an *additional* normalization that removes the initial year-length difference and isolates curvature. It is not the literal original calendar. Omitting the linear factor in the first equation would conflate them.

Published age [ka]	Literal Δu [years]	Local Δu [years]
100	-1.814	+0.488
200	-2.650	+1.953
300	-2.511	+4.395
420	-1.054	+8.614

Positive Δu means an older assigned coordinate age. Both columns are scenario outputs, not dating measurements.

The scenario curves in Figure 2 use these explicit maps and recompute constant-frequency fits in the resulting coordinates. If $v = g(u)$, a phase with frequency f_u has local coordinate frequency

$$f_v(u) = \frac{f_u}{g'(u)}. \tag{10}$$

This is a coordinate effect. If the complete orbital and climate phase models are transformed through $u = g^{-1}(v)$ consistently, the original physical predictions and sample correspondences are preserved. Fitting a new constant frequency in v instead uses a different basis. Reassigning the climate ages while keeping La2004 fixed would be a *redating hypothesis*, requiring independent age markers and their uncertainties.

5 How much the trial clocks change the frequency image

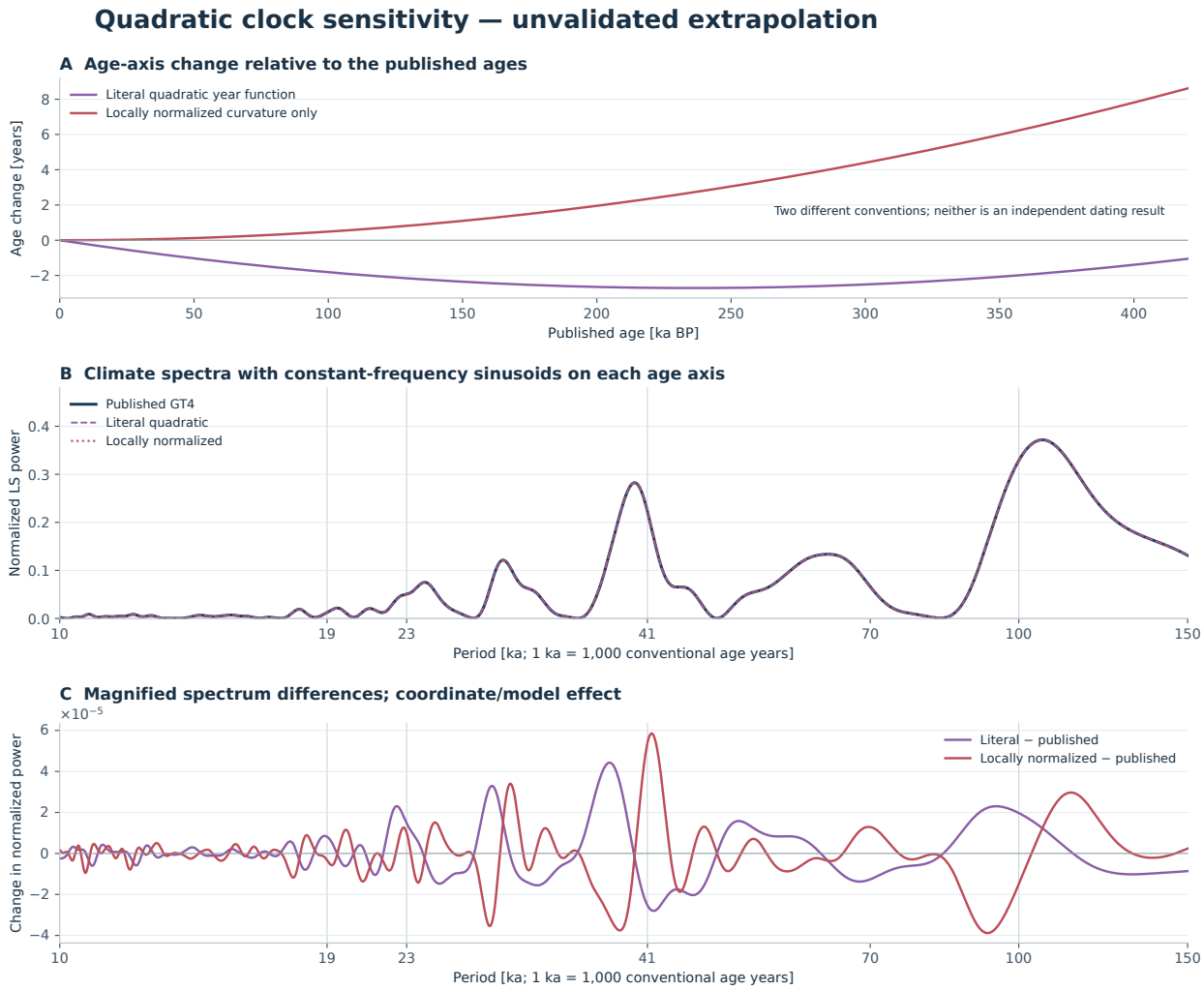


Figure 2: Unvalidated quadratic clock sensitivities, with the literal and locally normalized mappings kept separate. The lower comparison measures the effect of fitting a constant-frequency basis after each coordinate change; it does not measure an observed change in the Earth's history. The exact affine Quant conversion is the invariant reference.

6 What follows, and how to test more

Supported by this calculation. Published warm/cold variations can be compared with actual long-term orbital frequency patterns. The exact Quant conversion retains those patterns. The specific quadratic extrapolations produce the small changes tabulated above; they do not remove the major glacial bands or independently supply a different age of the Earth.

What the comparison does not identify. Frequency overlap alone does not determine an energy-transfer law, prove two cosmic streams, establish a preferred absolute frame or show that ordinary relative-motion measurements are wrong. Insolation is one input to a climate system with memory and feedback. An Antarctic isotope-derived temperature is not identical to northern solstice insolation, either in phase or amplitude. Standardizing curves for display does not make their units physically interchangeable.

A decisive next test. Freeze a candidate age map and a climate-response law before examining additional records. Compare them with the conventional model using independently dated ice, marine and terrestrial sequences; reserve full cycles or separate archives for validation. Include shared age errors, proxy uncertainty and calibration parameters. Fit any climate lag explicitly. An improvement obtained by moving age markers after inspecting the same spectrum is not independent evidence for a new clock.

Reproduction. The source package contains the downloaded NOAA and IMCCE files with checksums, `analyze.py`, the selected series and periodograms as CSV, numerical results as JSON, both figures, and this report's LaTeX source. Run `python3 analyze.py` with NumPy, SciPy and Matplotlib; then run `build.sh` for the PDF. Raw files are reused on subsequent runs.

References

[1] Petit, J.-R., et al. (2001 archive release). NOAA/WDS Paleoclimatology: Vostok isotope and gas data and temperature reconstruction. doi:10.25921/kcry-ae86. Data subset 0–420 ka BP; accessed 6 November 2026.

[2] Petit, J.-R., et al. (1999). Climate and atmospheric history of the past 420,000 years from the Vostok ice core, Antarctica. *Nature* 399, 429–436. doi:10.1038/20859.

[3] Laskar, J., et al. (2004). A long-term numerical solution for the insolation quantities of the Earth. *Astronomy & Astrophysics* 428, 261–285. doi:10.1051/0004-6361:20041335.

[4] IMCCE, Astronomical Solutions for Earth Paleoclimates. La2004 original data and README; accessed 6 November 2026. https://ssp.imcce.fr/insola/earth/online/earth/La2004/.

[5] Anders, J. C. (2026). *Uniquant: Zeit, Phase und Rekonstruktion*, working version 1.0.0. User-supplied manuscript, `UNIQUANT_Paper_V1_0_0(1).pdf`; not a peer-reviewed palaeoclimate time calibration. Equations and scenario conventions used here are stated explicitly above.